Specialist Physical AI Need Not Be Opaque
A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control, failure analysis, and weight-derived causal structure

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Abstract

Tensor decomposition turns a large table of learned interactions into simpler low-rank factors that can be ranked and inspected. Conventional robot policies contain operations that block this direct weight-level view. We introduce χ-VLA, a 20.1M-parameter vision-language-action policy built only from bilinear, linear, and rational learned operations. Its weights can therefore be reorganized into explicit interaction tensors rather than treated only as an opaque checkpoint. A mechanical audit and numerical reconstructions verify that the deployed policy preserves this form. Trained from scratch on LIBERO-Object, convolutional χ-VLA reaches $93.7\% \pm 0.8\%$ over three seeds under the published protocol. We reconstruct every attention module in three checkpoints with worst relative error 2.56×10^{-7}, and exact weight-derived subspaces beat equal-rank random controls in all 36 primary block-head tests. Across three checkpoints, retaining 64 of 384 causally ranked visual dimensions succeeds in $29/60$ matched trials, compared with $47/60$ for the full representation and $0/60$ for random subspaces. The causal effect replicates, but near-lossless preservation does not. On disjoint cache samples, the selected basis has at least $9.9\times$ lower reconstruction error than the median of five random projectors in every action group and checkpoint. A 300-pair instruction intervention reliably changes motion but rarely overcomes the scene prior completely, and a fifteen-configuration direct coefficient-edit sweep yields no candidate that passes both selectivity gates. A same-skeleton conventional twin differs by only 0.006% in parameters. Complete same-hardware replications find a 4.4-point rational deficit on A30 and 5.8 points on A6000. Compilation reduces the same-A30 latency ratio from 3.49–$5.33\times$ eager to 1.73–$1.74\times$ compiled. The result is a capable and structurally inspectable specialist, not a zero-shot generalist, a compute-matched system, or a solved editing interface.

1 Introduction

Modern robot policies can achieve high success while giving little indication of why an action was chosen. Their checkpoints contain millions of weights, but those weights do not directly show which image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated activations, yet a correlation is not automatically a mechanism or a safe editing target.

A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction table defined by its weights. Its decomposed factors then provide candidate control mechanisms that

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can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully tensor-decomposable policy.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

We study the narrower but testable question:

> *Can a specialist VLA retain competitive closed-loop capability when every deployed learned operation is polynomial or rational, and does the resulting structure provide a useful weight-based analysis surface?*

The stronger thesis asks for three results on the same policies: small capability cost, stable weight-derived causal structure, and selective edits without retraining. We establish the first two only partially and report the failed first surgery screen rather than treating algebraic representability as proof of interpretability.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%. A same-skeleton conventional twin differs by only 0.006% in parameters. Two complete same-hardware replications place the seed-0 capability cost between 4.4 and 5.8 points.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.
5. We show a causal closed-loop consequence. A selected 64-dimensional visual subspace consistently outperforms random equal-width subspaces across three checkpoints, while remaining materially below the full policy. A separate three-checkpoint instruction intervention distinguishes reliable trajectory response from incomplete task-level rerouting.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r\times(d+1)}$ and $D \in \mathbb{R}^{d\times r}$,

$$\mathrm{BFFN}(x) = D\left[(L\bar{x}) \odot (R\bar{x})\right], \qquad y_o = \sum_{i,j}\left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j}\right)\bar{x}_i\bar{x}_j. \tag{1}$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C\left[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top)/s\right] V. \tag{2}$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with interaction coefficients determined by the weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z+\epsilon}$:

$$\mathrm{RNorm}(x) = x\, r\left(\frac{1}{d}\sum_{j=1}^{d} x_j^2\right). \tag{3}$$

Here rational means a ratio of polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D. Updating this pair preserves input-specific rescaling without leaving the rational tensor representation.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has 20,137,352 learned parameters.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

The normalization reconstruction has maximum absolute error 1.64×10^{-6}. A bilinear FFN reconstruction has median relative error 1.40×10^{-7}. These residuals are bounded by the deployed module's float32 statistic. The audit certifies operator membership and local reconstruction, as summarized in Appendix Figure 9. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and 1,500 rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

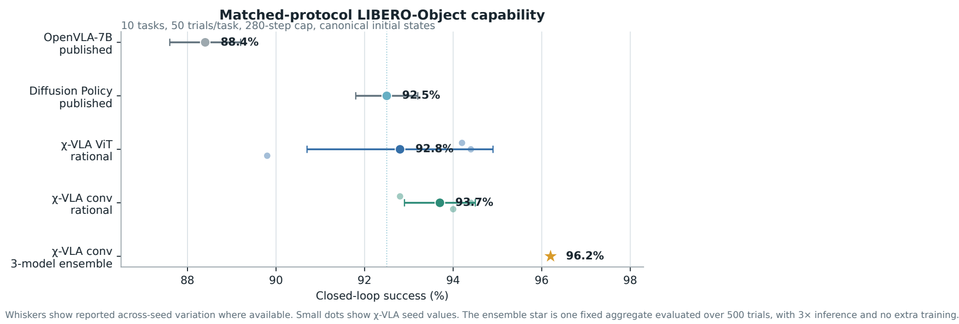

Figure 1: **Protocol-aligned LIBERO-Object capability.** Small dots are χ-VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ-VLA, rational ViT	20.1M	3	$92.8\% \pm 2.1\%$
Same-skeleton conventional ViT	20.1M	1	89.8%
χ-VLA, rational conv	20.1M	3	$\mathbf{93.7\% \pm 0.8\%}$
χ-VLA, conv ensemble	3×20.1M	$3 \to 1$	$\mathbf{96.2\%}$
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%, and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed comparison.

The same-skeleton conventional twin replaces the rational blocks with softmax attention, SwiGLU, and RMSNorm while holding the data, updates, seed, decoder, and 500 canonical trials fixed. On A30, the conventional twin succeeds in $448/500$ trials and the rational checkpoint in $426/500$. On A6000, they reach $451/500$ and $422/500$. Their parameter counts differ by only 0.006%. The rational deficit is 4.4 and 5.8 points, and task 3 accounts for 21 lost successes on each platform. Targeted replications reverse the task-3 gap: χ-VLA seeds 1 and 2 each reach $41/50$, compared with $37/50$ and $33/50$ for their conventional counterparts. The seed-0 task gap is therefore not directionally consistent across seeds. Full-suite seed 1 reaches $404/500$ for χ-VLA and $45/500$ for the conventional checkpoint, but different trainer provenance prevents a conversion-cost interpretation. The collapsed conventional seeds fit the training cache with normalized MSE of 9.51×10^{-4} and 1.19×10^{-3}, which rules out gross checkpoint corruption but not closed-loop covariate shift. The same χ-VLA checkpoint reaches $423/500$ on A40, between its A30 and A6000 results. On one A30, eager latency is 38.05 versus 7.14 ms. Compilation reduces it to 1.52 versus 0.88 ms, a $1.74\times$ gap. Rational training is $3.12\times$ slower at batch 128 and uses $1.63\times$ the peak allocated memory. A6000 profiling confirms a $1.73\times$ compiled ratio.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. Seed 0 misses the predeclared three-point noninferiority margin on both GPU classes. Conventional seeds 1 and 2 completed at only 9.0% and 8.2%, despite low final-minibatch losses and excellent training-cache fit. This exposes severe closed-loop seed sensitivity rather than a stable across-seed conversion cost. The weak checkpoints were trained natively on Athena, while the strong seed-0 checkpoint is a legacy artifact. A fully native matrix, matched χ-VLA evaluations, and partial-conversion controls remain necessary.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15}. The internal core identity agrees to 1.24×10^{-14}.

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16}. We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5}. This residual comes from the deployed implementation's intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors rank important directions without constructing the full table. It matches brute-force materialization to 2.13×10^{-14} at toy scale.

4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached activations are used only afterward to score how well it preserves the layer's computation.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure and a language shortcut

We next test whether the identified structure affects closed-loop behavior and whether high task success reflects robust use of the language input.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we rank directions separately for translation, rotation, and gripper actions. A subspace is a smaller coordinate system inside the model's 384-dimensional visual representation. Retaining it means zeroing all other directions, then running the actual policy. We compare selected and random equal-width subspaces. Because the ranking uses cached training demonstrations and downstream action gradients, offline reconstruction remains in-cache. Its evaluation samples are disjoint from the samples that estimate the ranking. Closed-loop intervention uses independent canonical simulator states, so it tests a causal bottleneck but is not a weight-only discovery result. An earlier one-checkpoint screen found in-cache random-to-selected reconstruction ratios of $4.8\times$, $4.6\times$, and $41.8\times$ for translation, rotation, and gripper, followed by $19/20$, $18/20$, and $0/20$ closed-loop successes for full, selected, and random conditions. Appendix Figure 4 reports the stronger replication. Across checkpoints, the three conditions reach $47/60$, $29/60$, and $0/60$. The selected-versus-random direction replicates with an exact paired sign $p < 0.008$ at every checkpoint, but the selected subspace remains 30 points below the full policy. The earlier near-lossless result does not replicate. Offline fidelity now uses $1{,}024$ discovery samples and $1{,}024$ disjoint evaluation samples per checkpoint. The selected basis has 16.2–$32.3\times$ lower translation error, 9.9–$15.0\times$ lower rotation error, and 15.4–$30.9\times$ lower gripper error than the median of five equal-rank random projectors. This repairs sample reuse but remains an in-cache diagnostic. The exact layerwise attention method in Section 4 must still be connected to the same full-suite closed-loop intervention.

5.2 Counterfactual instruction test

High success does not guarantee robust language use. A one-step screen responds to a renamed visible object in only 15.0% of 80 states. We therefore run 100 paired 80-step rollouts for each of three checkpoints. Relative end-effector preference for the renamed object increases by 0.175, 0.168, and 0.189 m, with all trial-bootstrap 95% intervals above zero. Motion shifts toward that object in 84%, 86%, and 88% of pairs, yet only 33%, 28%, and 28% end closer to it. Appendix Figure 3 shows the replication. The seed-0 conventional twin has a 0.239 m shift, 92% directional response, and the same 33% endpoint rate as rational seed 0. Language response is therefore not unique to tensor decomposability. This is a reach-phase proximity outcome, not counterfactual task success.

The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target, so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy range.

This mixed result narrows the VLA claim. Language changes the trajectory reproducibly, but scene priors often prevent complete rerouting. The model is a capable language-conditioned specialist, not yet a robust instruction-grounded policy. LIBERO-CF independently documents the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively repair that shortcut.

6 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ-nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and causal controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The same-skeleton seed-0 control has a 4.4 to 5.8 point capability gap and a residual compiled-runtime tax. Two additional conventional seeds collapse below 10%, so training stability remains unresolved. The benchmark is one specialist suite, counterfactual instructions do not fully reroute the policy, and exact decomposition is layerwise rather than end to end. The closed-loop subspace test replicates across three checkpoints but loses 18 of the full policy's 47 successes, while a fifteen-configuration discovery sweep yields no coefficient edit that passes both selectivity gates. The defensible result is therefore narrow: rational restrictions do not prevent competitive LIBERO-Object control, and they provide a precise weight-derived analysis surface. Broader grounding, multi-seed controls, optimized execution, and validated coefficient edits remain necessary.

Reproducibility. The final artifact will provide code, configurations, checkpoint hashes, exact certificates, and per-episode logs. Canonical-state checks fail loudly when requested states are unavailable.

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A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in $189/200$ trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

A.2 Matched architecture capability and systems cost

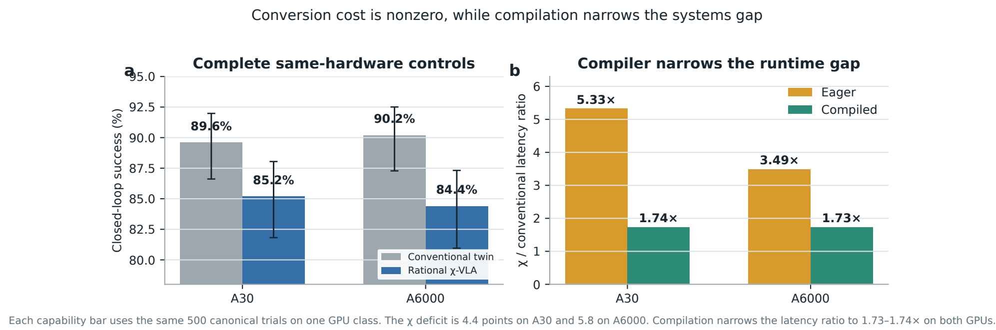

Figure 2: **Conversion cost is modest but nonzero, and compilation narrows the systems gap.** Complete same-hardware evaluations give 89.6% versus 85.2% on A30 and 90.2% versus 84.4% on A6000, conventional then rational. Parameters differ by 0.006%. Compilation narrows the batch-one latency ratio to 1.73–$1.74\times$ on both GPUs.

A.3 Three-checkpoint counterfactual grounding

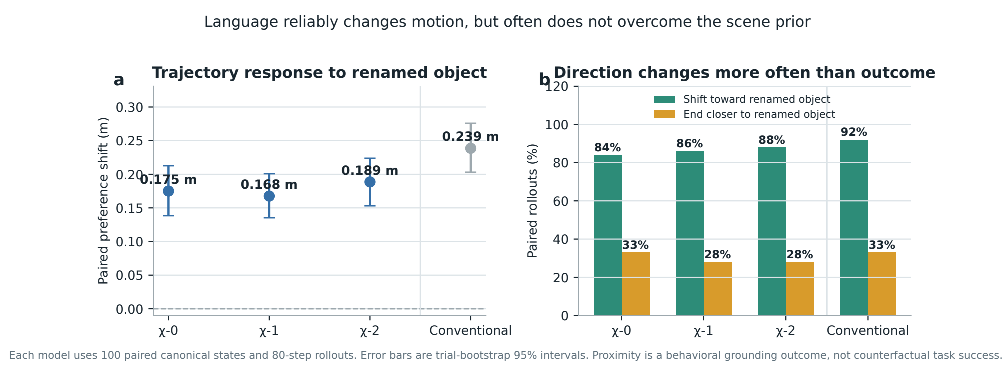

Figure 3: **Renaming the target changes motion reliably but does not fully override the scene prior.** Each model is evaluated on 100 paired canonical states for 80 control steps. The left panel reports the change in end-effector preference for the newly named object, with 95% bootstrap intervals. The right panel separates directional response from the stricter condition of ending closer to that object. These are proximity tests, not counterfactual task-success measurements. The conventional control shows that ordinary language response is not unique to tensor decomposability.

A.4 Causal visual-subspace intervention

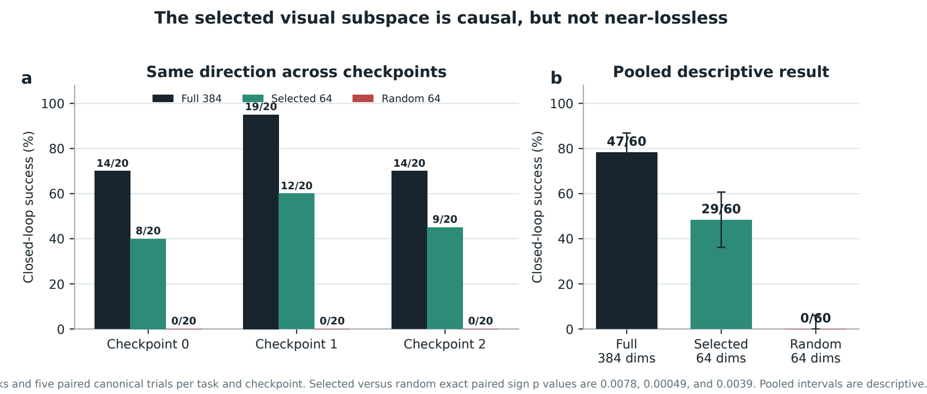

Figure 4: **The selected visual subspace has a replicated causal effect, but it is not near-lossless.** Each checkpoint uses four tasks and five paired canonical trials per task. Selected subspaces beat random controls in all three checkpoints. The pooled interval is descriptive because trials within a checkpoint share learned weights.

A.5 Per-task ensemble behavior

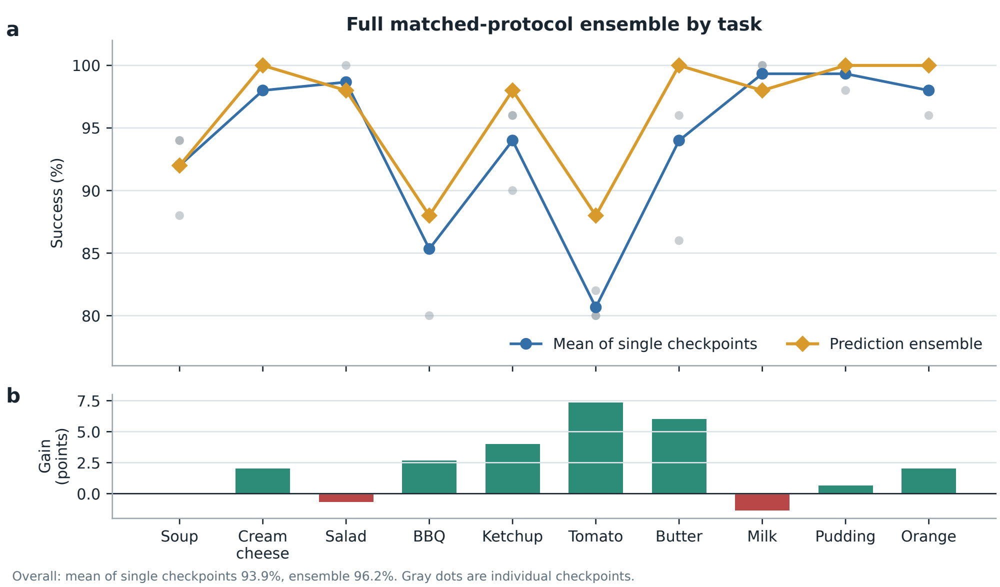

Figure 5: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

A.6 Product-routing multimodal head

For an action-query state h, the product-routing head represents

$$a(h, b) = c_0(h) + \sum_{g=1}^{G} (2b_g - 1)c_g(h), \qquad b \in \{0, 1\}^G. \tag{4}$$

The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$ and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head. Product routing remains a construction result until it wins a controlled genuinely multimodal task.

A.7 Development interventions

Focused DAgger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven corrective episodes and $1{,}131$ frames. Applying the same update thinly across all ten tasks reduces overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points. A reward audit shows that accumulated step cost makes every successful trajectory have negative return. After removing that term and repairing the terminal boundary, one pilot improves by two points. These are development diagnostics, not validated paper contributions.

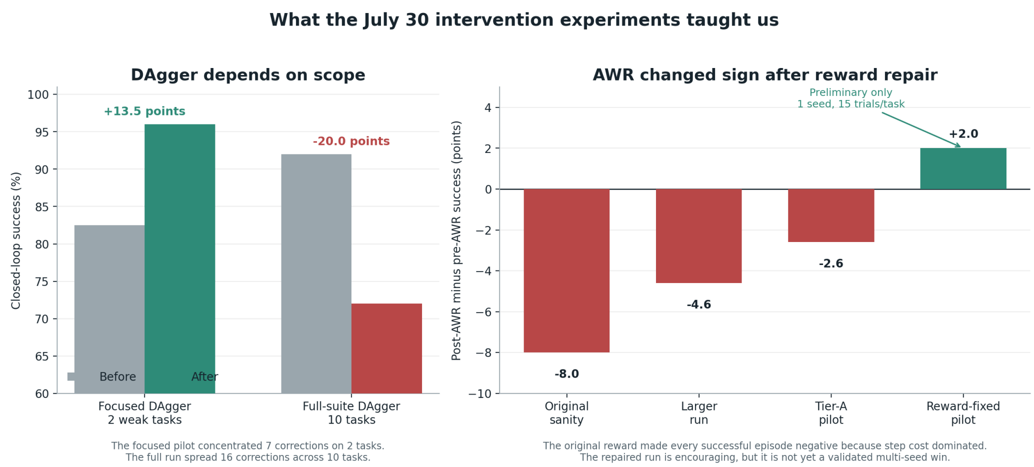

Figure 6: **Development interventions reveal scope and reward sensitivity.** Focused DAgger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

A.8 Expanded attention screen and first direct edit

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, $83/96$ block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only $54/96$ and $60/96$ pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-subspace retention preserves $18/20$ successes compared with $16/20$ for both random controls. This 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves $18/20$, compared with $17/20$ for both random controls, which is the wrong direction for the necessity prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and value column edits.

A subsequent discovery sweep evaluates blocks 0, 2, 4, 6, and 7 at ranks 64, 128, and 256. None of the fifteen configurations passes both prespecified 15-point gates. The best configuration, block 0 at rank 128, has a 10-point retention advantage and a 40-point removal advantage over Frobenius-matched random edits. It therefore does not advance to held-out confirmation. The selected keep and removal operations change 56.4% and 82.6% of the affected query, key, and value weights in relative Frobenius norm. This is a global functional ablation, not a small semantic edit.

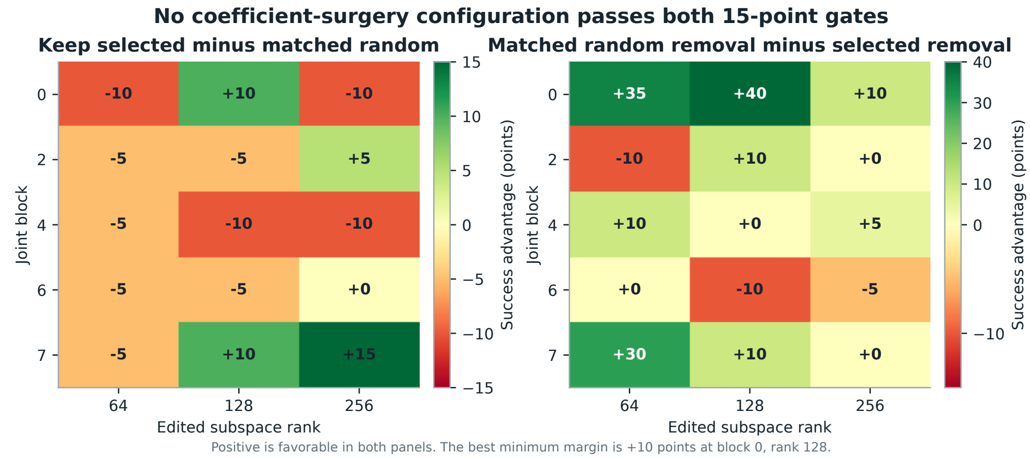

Figure 7: **The complete coefficient-surgery discovery sweep is negative.** Positive values favor the selected weight-derived subspace. No block-rank pair reaches the prespecified 15-point margin in both retention and removal.

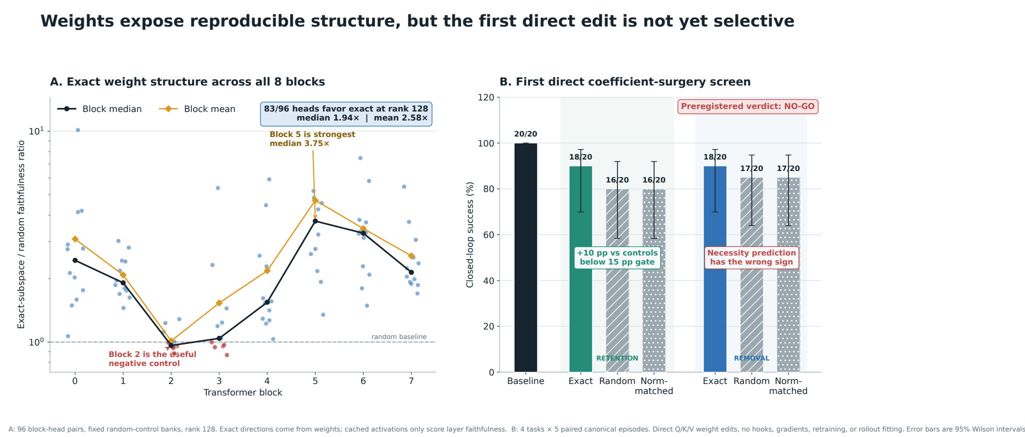

Figure 8: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

A.9 Numerical scope of rational conversion

The deployed Padé computation is exactly rational even though it approximates RMS normalization. Final certification must report its polynomial degrees, coefficients, denominator margin, observed activation range, tail error, action drift, and runtime overhead. Operator membership alone does not establish practical numerical safety under distribution shift.

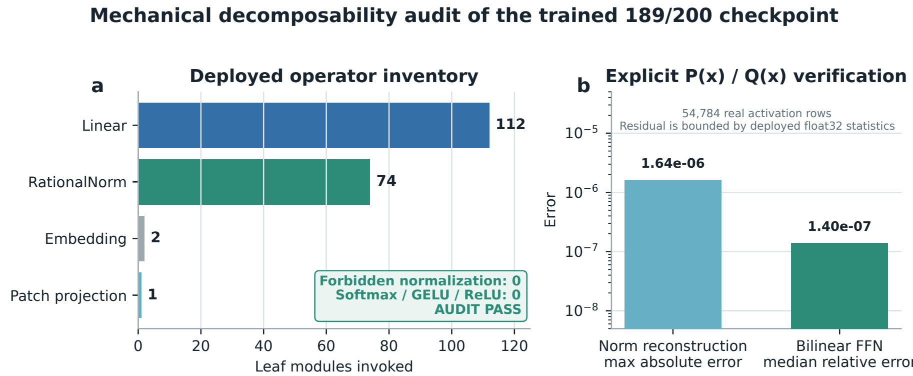

Figure 9: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.